

Table 10. SPI/I2S implementation

SPI features ⁽¹⁾	SPI1	SPI2
Hardware CRC calculation	X	X
Rx/Tx FIFO	X	X
NSS pulse mode	X	X
I ² S mode	X	-
TI mode	X	X

1. X = supported.

3.20 Universal serial bus device (USB) and host (USBH)

The devices embed a USB controller with full-speed USB device and host functionality, compliant with the USB specification version 2.0. The internal USB PHY supports USB FS signaling, embedded DP pull-up and also battery charging detection according to Battery Charging Specification Revision 1.2. The USB interface implements a full-speed (12 Mbit/s) function interface with added support for USB 2.0 Link Power Management. It has software-configurable endpoint setting with packet memory of 2 KB and suspend/resume support. It requires a precise 48 MHz clock coming from a high-speed external clock (from an external source or an oscillator) or generated by the internal 48 MHz oscillator in automatic trimming mode. The synchronization for this oscillator can be taken from the USB data stream itself (SOF signalization), which allows crystal-less operation in USB device mode.

3.21 Development support

3.21.1 Serial wire debug port (SW-DP)

An Arm SW-DP interface is provided to allow a serial wire debugging tool to be connected to the MCU.